ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

(i) f_x = 16/5 = 3.2 pixels, using focal length 16 mm and pixel size 5 microm. (ii) Sensor width is 6.4 mm, so field width = 400x6.4/16 = 160 mm and scale = 0.125 mm/pixel.

Q1(b) [3 marks]

ADC step = 5/2^12 = 1.2207 mV. Since sensitivity is 10 mV/N, ΔF = 1.2207/10 = 0.122 N. I report this as 0.122 mN.

Q1(c) [3 marks]

$$F = 0.8187, G = 0.0906$$

Question 2 [10 marks]

Q2(a) [2 marks]

Choose x_1 = θ and x_2 = $\dot{\theta}$. Then $\dot{x}_1 = x_2$ and $\dot{x}_2$ = [τ - b x_2 - mgℓ sin(x_1)]/J. At equilibrium:

$$x_2 = 0 and_0 = mg\ell sin\theta_e$$

Since |sin θ_e| less than or equal to 1, an equilibrium exists when $|\tau_0|$ less than or equal to mgℓ = 2.4525 N m.

At rest $\dot{\theta} = 0$ and $\ddot{\theta} = 0$, so the damping term vanishes. Rearranging gives $\tau_0 = mg\ell \sin\theta_e$. Here $mg\ell = 1 \times 9.81 \times 0.25 = 2.4525$ N m. Since $-1 \leq \sin\theta_e \leq 1$, the required torque must lie between -2.4525 and $+2.4525$ N m.

Q2(b) [2 marks]

$$A = [[0, 1], [-mg\ell sin(60^\circ)/J, -b/J]] = [[0, 1], [-106.2, -2.5]],$$
$$B = [[0], [50]]$$

Q2(c) [4 marks]

At rest, $\dot{\theta} = \ddot{\theta} = 0$, so the torque balance is Kp(θd-θss) = mgℓ sin θss. With Kp = 50, θd = π/2 and mgℓ = 2.4525, solve 50(π/2-θss) = 2.4525 sin θss. This gives θss approximately 1.52181 rad = 87.193°, so the error is 90°-87.193° = 2.807°; it is not zero.

Rearranging the steady-state equation gives $50(\pi/2 - \theta_{ss}) = 2.4525 \sin\theta_{ss}$. I used radians in the equation. Solving numerically gives $\theta_{ss} \approx 1.522$ rad $\approx 87.19°$, so the gap from $90°$ is about $2.81°$.

Q2(d) [2 marks]

With gravity compensation, the gravity term cancels the plant's mgℓ sin θ term. At rest, Kp(θd-θss)=0, so θss=θd and the steady-state error is zero. The linearised gravity stiffness term is also cancelled, so the closed-loop poles generally change; the stiffness contribution mgℓ cos θd is removed.

The compensation term cancels the nominal gravity torque in the plant model. At steady state the proportional term must then be zero, which requires $\theta_d - \theta_{ss} = 0$. This does not mean the transient response is unchanged: removing the gravity derivative changes the linearized stiffness.

Question 3 [10 marks]

Q3(a) [2 marks]

Finite differences are poor mainly because the encoder measures angle in degrees instead of radians, and because the velocity signal is always delayed by one full second.

Q3(b) [2 marks]

For H(s)=ωc/(s+ωc), ωc=2π(20)=125.664 rad/s and T=0.001 s. Tustin gives H(z)=α(1+z^{-1})/[(1+α)+(α-1)z^{-1}], α=ωcT/2=0.0628319. Thus:

$$y[k] = 0.881765y[k-1] + 0.0591175(x_k + x[k-1])$$

DC gain = (2x0.0591175)/(1-0.881765) approximately 1.000.

$\omega_c = 2\pi(20) = 125.664$ rad/s. With $T = 0.001$ s, $\alpha = \omega_c T/2 = 125.664(0.001)/2 = 0.062832$. The recurrence coefficients are approximately 0.881765 on the previous output and 0.0591175 on each input sample. Check: $2(0.0591175)/(1 - 0.881765) \approx 1$, so the DC gain is close to one.

Q3(c) [4 marks]

A-LC=[[-ℓ_1,1], [-ℓ_2,-2]]. I obtain p(s)=s^2+(ℓ_1+2)s+(ℓ_1+ℓ_2). Matching (s+10)^2 gives ℓ_1=18 and ℓ_2=82.

Q3(d) [2 marks]

An observer can estimate velocity from position measurements. A filter can be safer because it smooths noise.